This argument works equally well over any open subset of $B_n - B_{n-1}$ that is path-connected, in particular over $U \cap (B_n - B_{n-1})$, so via Lemma 4K.3(a) this finishes the proof that $SP(A) \to SP(X) \to SP(X/A)$ is a quasifibration.

Since the homotopy axiom is obvious, this gives us the first two of the three axioms needed for the groups $h_i(X) = \pi_i SP(X)$ to define a reduced homology theory. There remains only the wedge sum axiom, $h_i(\bigvee_\alpha X_\alpha) \approx \bigoplus_\alpha h_i(X_\alpha)$, but this is immediate from the evident fact that $SP(\bigvee_\alpha X_\alpha) = \prod_\alpha SP(X_\alpha)$, where this is the 'weak' product, the union of the products of finitely many factors.

The homology theory $h_*(X)$ is defined on the category of connected, basepointed simplicial complexes, with basepoint-preserving maps. The coefficients of this homology theory, the groups $h_i(S^n)$, are the same as for ordinary homology with $\mathbb{Z}$ coefficients since we know this is true for $n = 2$ by the homeomorphism $SP(S^2) \approx \mathbb{C}\mathrm{P}^\infty$, and there are isomorphisms $h_i(X) \approx h_{i+1}(\Sigma X)$ in any reduced homology theory. If the homology theory $h_*(X)$ were defined on the category of all simplicial complexes, without basepoints, then Theorem 4.59 would give natural isomorphisms $h_i(X) \approx H_i(X;\mathbb{Z})$ for all X, and the proof would be complete. However, it is easy to achieve this by defining a new homology theory $h'_i(X) = h_{i+1}(\Sigma X)$, since the suspension of an arbitrary complex is connected and the suspension of an arbitrary map is basepoint-preserving, taking the basepoint to be one of the suspension points. Since $h'_i(X)$ is naturally isomorphic to $h_i(X)$ if X is connected, we are done. $\square$

It is worth noting that the map $\pi_i(X) \to \pi_i SP(X) = H_i(X;\mathbb{Z})$ induced by the inclusion $X = SP_1(X) \hookrightarrow SP(X)$ is the Hurewicz homomorphism. For by definition of the Hurewicz homomorphism and naturality this reduces to the case $X = S^i$, where the map $SP_1(S^i) \hookrightarrow SP(S^i)$ induces on π_i a homomorphism $\mathbb{Z} \to \mathbb{Z}$, which one just has to check is an isomorphism, the Hurewicz homomorphism being determined only up to sign. The suspension isomorphism gives a further reduction to the case $i = 1$, where the inclusion $SP_1(S^1) \hookrightarrow SP(S^1)$ is a homotopy equivalence, hence induces an isomorphism on π_1.

Exercises

1. Show that Corollary 4K.2 remains valid when X and Y are CW complexes and the subspaces U_i and V_i are subcomplexes rather than open sets.

2. Show that a simplicial map $f : K \to L$ is a homotopy equivalence if $f^{-1}(x)$ is contractible for all $x \in L$. [Consider the cover of L by open stars of simplices and the cover of K by the preimages of these open stars.]

3. Show that $SP_n(I) = \Delta^n$.

4. Show that $SP_2(S^1)$ is a Möbius band, and that this is consistent with the description of $SP_2(S^n)$ as a mapping cone given in Example 4K.5.

5. A map $p : E \to B$ with B not necessarily path-connected is defined to be a quasifibration if the following equivalent conditions are satisfied:

(i) For all $b \in B$ and $x_0 \in p^{-1}(b)$, the map $p_* : \pi_i(E, p^{-1}(b), x_0) \to \pi_i(B, b)$ is an isomorphism for $i > 0$ and $\pi_0(p^{-1}(b), x_0) \to \pi_0(E, x_0) \to \pi_0(B, b)$ is exact.

(ii) The inclusion of the fiber $p^{-1}(b)$ into the homotopy fiber F_b of p over b is a weak homotopy equivalence for all $b \in B$.

(iii) The restriction of p over each path-component of B is a quasifibration according to the definition in this section.

Show these three conditions are equivalent, and prove Lemma 4K.3 for quasifibrations over non-pathconnected base spaces.

6. Let X be a complex of spaces over a Δ-complex Γ, as defined in §4.G. Show that the natural projection $\Delta X \to \Gamma$ is a quasifibration if all the maps in X associated to edges of Γ are weak homotopy equivalences.

4.L Steenrod Squares and Powers

The main objects of study in this section are certain homomorphisms called **Steenrod squares** and **Steenrod powers**:

$$Sq^i : H^n(X; \mathbb{Z}_2) \to H^{n+i}(X; \mathbb{Z}_2)$$
$$P^i : H^n(X; \mathbb{Z}_p) \to H^{n+2i(p-1)}(X; \mathbb{Z}_p) \quad \text{for odd primes } p$$

The terms 'squares' and 'powers' arise from the fact that Sq^i and P^i are related to the maps $\alpha \mapsto \alpha^2$ and $\alpha \mapsto \alpha^p$ sending a cohomology class α to the 2-fold or p-fold cup product with itself. Unlike cup products, however, the operations Sq^i and P^i are stable, that is, invariant under suspension.

The operations Sq^i generate an algebra $\mathcal{A}_2$, called the Steenrod algebra, such that $H^*(X; \mathbb{Z}_2)$ is a module over $\mathcal{A}_2$ for every space X, and maps between spaces induce homomorphisms of $\mathcal{A}_2$-modules. Similarly, for odd primes p, $H^*(X; \mathbb{Z}_p)$ is a module over a corresponding Steenrod algebra $\mathcal{A}_p$ generated by the P^i's and Bockstein homomorphisms. Like the ring structure given by cup product, these module structures impose strong constraints on spaces and maps. For example, we will use them to show that there do not exist spaces X with $H^*(X; \mathbb{Z})$ a polynomial ring $\mathbb{Z}[\alpha]$ unless α has dimension 2 or 4, where there are the familiar examples of $\mathbb{C}P^\infty$ and $\mathbb{H}P^\infty$.

This rather lengthy section is divided into two main parts. The first part describes the basic properties of Steenrod squares and powers and gives a number of examples and applications. The second part is devoted to constructing the squares and powers and showing they satisfy the basic properties listed in the first part. More extensive